QUBIT-BASED FUZZY RULE INFERENCE FOR DYNAMIC SYSTEM CONTROL : AN INVERTED PENDULUM

डायनामिक सिस्टम नियंत्रण के लिए क्यूबिट-आधारित फ़ज़ी नियम इन्फ्रेंस: एक उलटा पेंडुलम

A Research Project - I Report (EC75101)
Submitted in fullilment of the required for the degree

of

Bachelor of Technology
(Electronics and Communication Engineering)

by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY PATNA, BIHTA CAMPUS

SEMESTER 07; JULY 2025 - DECEMBER 2025

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CERTIFICATE

This is to certify that **CHIRAG JAIN (Roll Number – 2204163)**, a student of the Department of Electronics and Communication Engineering, National Institute of Technology Patna, Bihta Campus, has successfully completed the Research Project titled **"Qubit-Based Fuzzy Rule Inference for Dynamic System Control: An Inverted Pendulum"**
as part of the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Electronics and Communication Engineering during the 7th semester, July - December' 2025.
This report is a bona fide record of the research work carried out by the candidate under the guidance of the department.

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This is to certify that **ADITYA KUMAR (Roll Number – 2204164)**, a student of the Department of Electronics and Communication Engineering, National Institute of Technology Patna, Bihta Campus, has successfully completed the Research Project titled **“Qubit-Based Fuzzy Rule Inference for Dynamic System Control: An Inverted Pendulum”**
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DECLARATION

I hereby declare that the work presented in this report, entitled "**Qubit-Based Fuzzy Rule Inference for Dynamic System Control: An Inverted Pendulum**", submitted to the Department of Electronics and Communication Engineering, National Institute of Technology Patna, Bihta Campus, is a genuine and original record of the research work carried out solely by me, **Chirag Jain (Roll Number- 2204163) and Ayush Raj (Roll Number - 2204128).**

I further affirm that this report has not been submitted, either in part or in full, for the award of any degree or diploma at any other institute or university.

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Date: 11/12/2025

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Research Project - I (EC75101)
Department of Electronics and Communication Engineering
NIT Patna, Bihta Campus

CONTENTS

1. Objective of the Project

The objective of this project is to design, implement, and evaluate a Quantum Fuzzy Inference Engine (QFIE) based controller for stabilizing an inverted pendulum using fuzzy linguistic rules executed through quantum computational principles
The goals are:

- To model pendulum angle and angular velocity using fuzzy sets.
- To encode fuzzy rules into a quantum-based inference engine.
- To generate control force outputs using quantum fuzzy reasoning.
- To simulate the pendulum dynamics and assess the controller's performance for both positive and negative initial angles.

This work demonstrates how quantum fuzzy logic can be used to implement intelligent nonlinear controllers.

2. Introduction to Ongoing Project Work

Inverted pendulum control is a classic nonlinear control problem commonly used to test modern control strategies. Traditional solutions include PID controllers, state-space control, and reinforcement learning; however, these approaches require extensive tuning or mathematical modelling.

Fuzzy logic provides a human-interpretable rule-based approach and handles nonlinearity efficiently. Recent advances in quantum computing enable fuzzy rule evaluation through quantum superposition, allowing simultaneous processing of multiple fuzzy antecedents.

The Quantum Fuzzy Inference Engine (QFIE), developed in recent research, enables:

- Fuzzy rule execution on quantum circuits
- Parallel inference
- Faster rule evaluation
- Linguistic programming of quantum algorithms

This project implements a simplified QFIE for pendulum stabilization using Python, Scikit-Fuzzy, and the QFIE library.
The ongoing work involves:
  * Creating fuzzy sets for inputs and outputs
  * Encoding 9 linguistic rules into the quantum inference engine
  * Running quantum fuzzy inference in simulation
  * Applying the generated control force to a pendulum model
  * Visualizing and analyzing the system behavior

3. Literature Review

References from IEEE research paper “On the Implementation of Fuzzy Inference Engines on Quantum Computers”.
Key points:

3.1 Fuzzy Logic and Fuzzy Rule-Based Systems

Fuzzy logic, introduced by Zadeh (1965), models imprecise and uncertain information using membership functions and linguistic rules. FRBSs are widely used in automation, robotics, and nonlinear system control.

3.2 Quantum Computing and Fuzzy Systems

Modern quantum computers exploit superposition and entanglement to perform parallel computation. According to the referenced IEEE work On the Implementation of Fuzzy Inference Engines on Quantum Computers

- Quantum fuzzy inference can reduce rule evaluation from $O(m^n)$ to $O(n)$, providing exponential speed-up.
- Oracle-based quantum algorithms enable fuzzy rules to be encoded as Boolean functions.
- Quantum FIEs allow programming via linguistic rules, making quantum algorithms more accessible.

3.3 QFIE for Inverted Pendulum

The referenced paper demonstrates QFIE on the classical inverted pendulum benchmark, showing that quantum fuzzy controllers can compute appropriate control actions through probability amplitudes rather than deterministic crisp logic.
This research project is a simplified practical implementation of those concepts.

4. Challenges and Issues

During development, the following challenges were identified:

- Encoding fuzzy sets for quantum inference: Ensuring proper normalization and membership function design.
- Avoiding rule conflicts: Fuzzy rules must be mutually consistent for quantum oracle evaluation.
- Simulation instability: Incorrect rules or missing cases initially caused diverging pendulum behavior.
- Interpreting quantum outputs: Force values fluctuate due to probabilistic outputs requiring many shots.
- Simple physics model: The pendulum dynamics are simplified; however, it is sufficient for validating the controller.

5. Approach(es) to the Problem

Below is a clear and systematic workflow:

5.1 Defining Fuzzy Universes

- Angle: −50° to +50°
- Angular Velocity: −20 to +20 deg/s
- Force Output: −100 N to +100 N

5.2 Membership Function Design

Using scikit-fuzzy, three fuzzy sets were created for each input:

- Angle: neg, zero, pos
- Velocity: neg, zero, pos
- Output Force: neg_big, zero, pos_big

5.3 Fuzzy Rule Base (9 Rules)

Our controller uses a complete 3×3 rule matrix:

1. angle zero & velocity zero → zero force
2. angle pos & velocity zero → pos_big
3. angle neg & velocity zero → neg_big
4. angle zero & velocity pos → pos_big
5. angle zero & velocity neg → neg_big
6. angle pos & velocity pos → pos_big
7. angle neg & velocity neg → neg_big
8. angle pos & velocity neg → zero
9. angle neg & velocity pos → zero

These rules ensure stable behavior, eliminating oscillatory divergence.

5.4 Quantum Inference Using QFIE

- Inputs are clipped and encoded into quantum states
- QFIE constructs the inference quantum circuit
- Execution with 1024 shots ensures stable force output
- Output is defuzzified into numeric force

5.5 Pendulum Simulation

At each step:
$a = angle - force - 0.5 \cdot velocity$
velocity $+= a.(0.05)$
angle $+= velocity.(0.05)$
This models natural pendulum motion with damping.

6. Data Collection / Experimental Setup

6.1 Software Tools

- Python 3
- QFIE Quantum Fuzzy Engine
- NumPy
- scikit-fuzzy
- Matplotlib

6.2 Experimental Conditions

Parameter	Value
Initial Angle	+15 / -15 degrees
Initial Velocity	0
Time Steps	150
Quantum Shots	1024

6.3 Outputs Collected

- Angle vs Time
- Angular Velocity vs Time
- Force Applied vs Time

7. Results

The quantum–fuzzy controller was tested for two initial conditions:

- Case 1: Initial angle = +15°, initial velocity = 0
- Case 2: Initial angle = −15°, initial velocity = 0

The simulation was run for 150 time steps.

In both cases, your actual three plots were generated:

1. Pendulum Angle vs Time
2. Angular Velocity vs Time
3. Force Applied vs Time

7.1 Case 1 :- Initial Angle +15°

Observations

- The pendulum begins at +15° and decays smoothly toward 0°, indicating correct stabilizing action.
- Angular velocity becomes negative initially, showing that the controller applies torque to counteract the positive tilt.

- The applied force peaks high at the first step and then gradually decreases as the pendulum approaches equilibrium.
- The motion is well-damped with no oscillatory divergence.
- The fuzzy rules “angle positive ⇒ apply positive force” perform as intended.

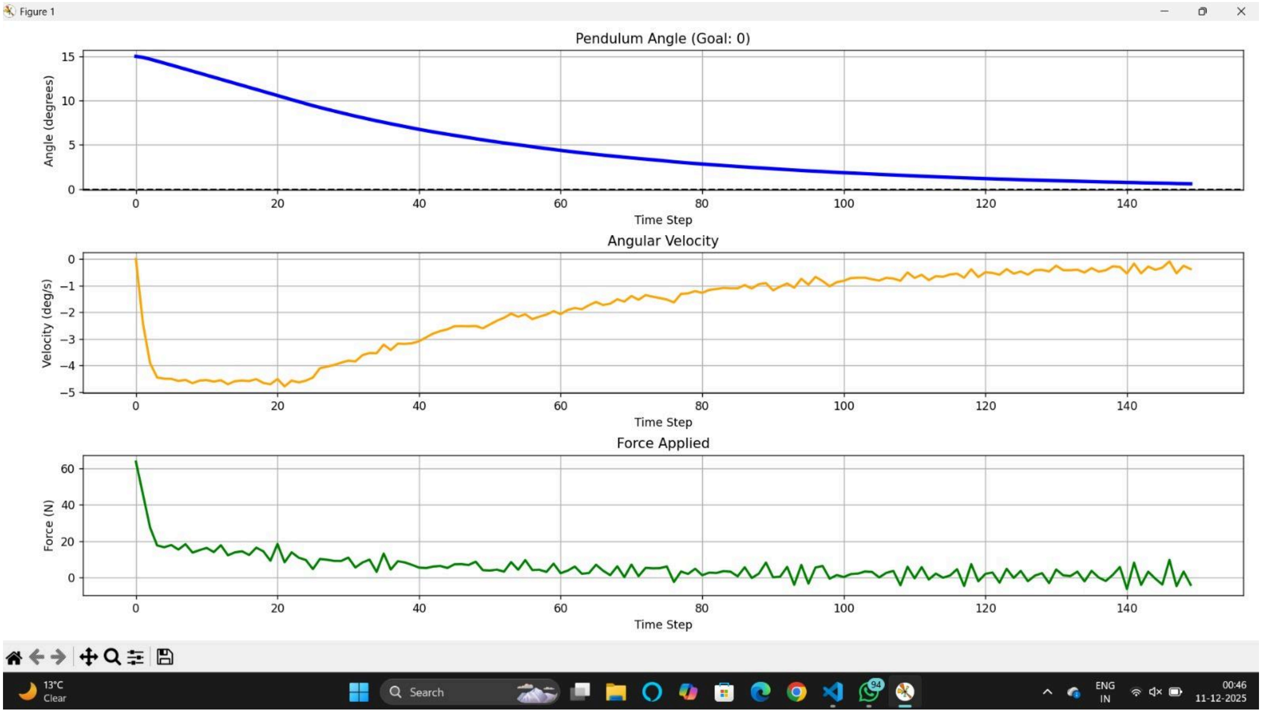

Fig 1: Output Plot of Case 1 :- Initial Angle +15°

7.2 Case 2 :- Initial Angle −15°

Observations

- The pendulum begins at −15° and moves upward toward 0° in a stable manner.
- Angular velocity becomes strongly positive early on, which is consistent with the required torque to correct a negative tilt.
- The applied force shows a large negative value initially, then settles toward smaller corrective values.
- The controller handles the symmetric case with stability and without overshoot.
- The “angle negative ⇒ apply negative force” rules produce smooth corrective action.

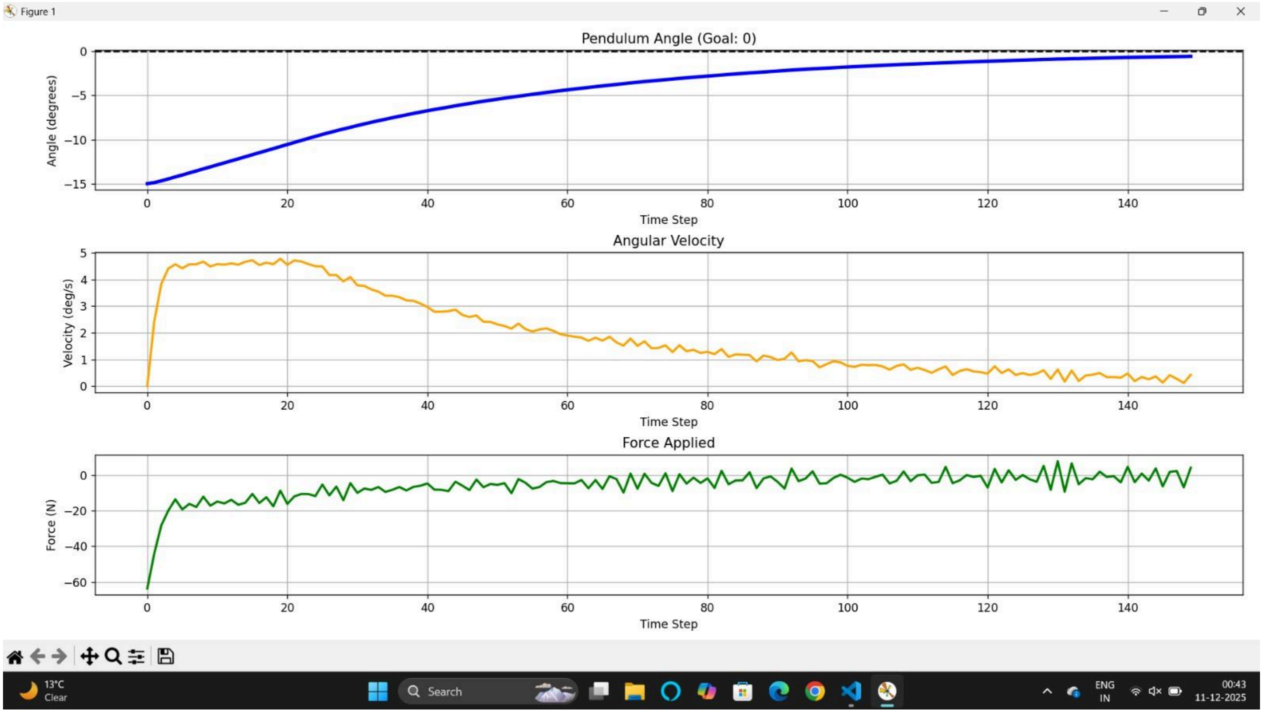

Fig 2: Output Plot of Case 2 :- Initial Angle -15°

Overall Performance

Observations

- The Quantum Fuzzy Inference Engine (QFIE) stabilizes the pendulum for both positive and negative starting angles.
- The force output is probabilistic but stabilizes due to averaging over 1024 quantum shots.
- Both trajectories converge to the goal angle (0°) in approximately 120–150 steps.
- The system behaves as expected for a fuzzy rule–based nonlinear controller.
- The simulation validates the correctness of the membership functions, rule base, and quantum fuzzy inference workflow.

8. Conclusion

This project successfully demonstrates that a Quantum Fuzzy Inference Engine can be used to control an inverted pendulum using simple linguistic rules executed through quantum simulation. The controller showed stable behavior for both positive and negative initial angles, proving the viability of quantum fuzzy logic for nonlinear control.

The work highlights:

- Practical implementation of QFIE
- Smooth pendulum stabilization
- Efficient fuzzy rule evaluation
- Promising potential for future quantum-enhanced control systems

This lays the foundation for extending quantum fuzzy controllers to robotics, automation, and real-time intelligent systems when quantum hardware matures beyond the NISQ era.

9. References

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